Electromagnetic Theory & Quantum Physics

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

8 lecture sections · 15 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; 15 graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

1. Light carries both a wave scale and an energy packet
2. Luminosity is not flux
3. A blackbody is a thermal reference, not a colour label
4. Two shortcuts with different jobs
5. Counting photons
6. The photoelectric threshold
7. Radiation pushes, and dust feels it differently
8. Choose the measurement before the formula
9. Lecture + Problems: predict, calculate, check
10. Further practice and provenance

1. Light carries both a wave scale and an energy packet

Radio waves, visible light and gamma rays are the same kind of electromagnetic phenomenon. They differ in wavelength and frequency. In vacuum,

$$c = \lambda f, \qquad E_\gamma = hf = \frac{hc}{\lambda}. \tag{1}$$

A convenient conversion is $E(\mathrm{eV}) = 1239.84/\lambda(\mathrm{nm})$. A blue photon carries more energy than a red photon. But a bright red lamp can deliver more total power than a faint blue lamp because it sends more photons.

Use $c = 2.99792458 \times 10^{8}\ \mathrm{m\,s^{-1}}$, $h = 6.62607015 \times 10^{-34}\ \mathrm{J\,s}$, $1\ \mathrm{eV} = 1.602176634 \times 10^{-19}\ \mathrm{J}$, $\sigma = 5.670374419 \times 10^{-8}\ \mathrm{W\,m^{-2}\,K^{-4}}$, $R_\odot = 6.957 \times 10^{8}\ \mathrm{m}$ and $L_\odot = 3.828 \times 10^{26}\ \mathrm{W}$.

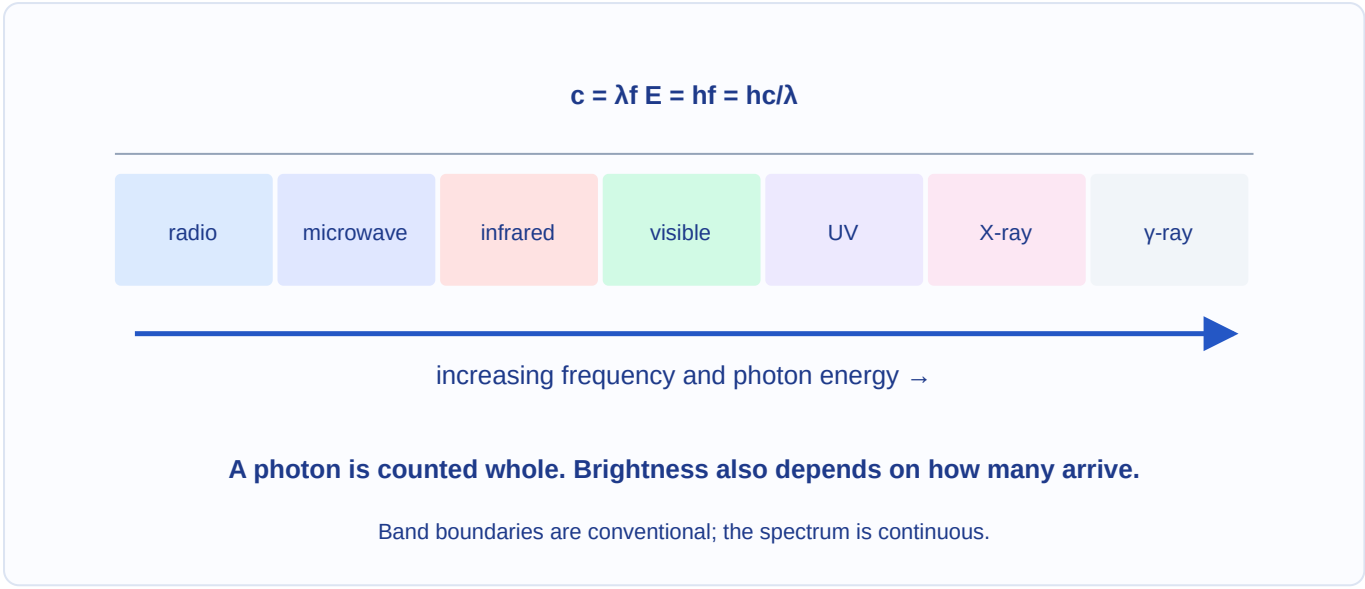


Figure 4. Wavelength tells you the energy of each photon, not the total luminosity. A detector's area, exposure and efficiency determine how many photons it records.

2. Luminosity is not flux

A star's luminosity L is the energy it emits per second. The flux F at a distance d is the power crossing each square metre of a sphere around it:

$$F = \frac{L}{4\pi d^2}. \tag{2}$$

The inverse square is geometry: the same power spreads over an area growing as d^2. Flux is not surface brightness, and neither is luminosity. A telescope collects power FA through area A, then loses some to throughput and detector efficiency.

3. A blackbody is a thermal reference, not a colour label

An ideal blackbody absorbs incident radiation and emits a spectrum determined only by temperature. Real stars are approximate blackbodies with spectral lines and wavelength-dependent opacity. Planck's law per unit wavelength is

$$B_\lambda(T) = \frac{2hc^2}{\lambda^5}\frac{1}{e^{hc/(\lambda k_B T)} - 1}. \tag{3}$$

It is a surface brightness per steradian per wavelength interval. You rarely need to evaluate the full expression in an Olympiad problem: Wien's and Stefan–Boltzmann's laws extract the peak and total power.

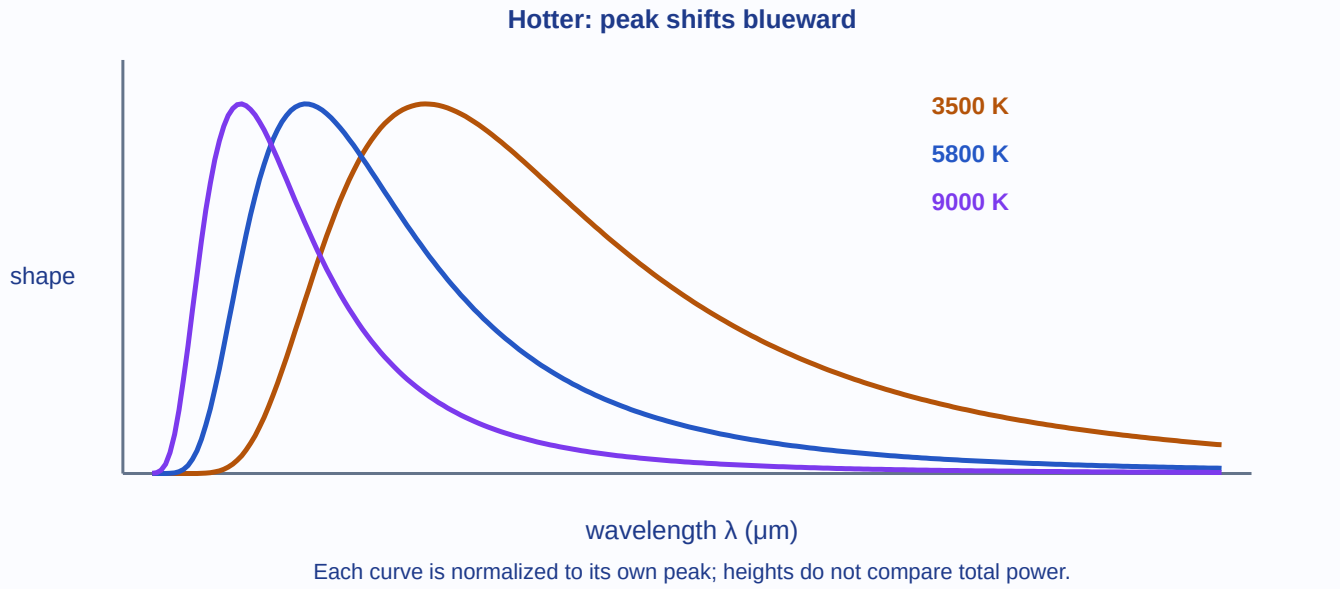


Figure 3. Wien's law locates the peak of the wavelength spectrum. Stefan–Boltzmann's law gives the total surface flux; the normalized heights here are not flux measurements.

4. Two shortcuts with different jobs

Wien's displacement law for the wavelength spectrum is

$$\lambda_{\max} T = 2.897771955 \times 10^{-3}\ \mathrm{m\,K}. \tag{4}$$

The Stefan–Boltzmann law integrates over all wavelengths:

$$F_{\mathrm{surface}} = \sigma T^4, \qquad L = 4\pi R^2 \sigma T^4. \tag{5}$$

A hotter star is bluer and emits much more power per unit area. A larger star can be cooler yet more luminous. The peak of B_λ is not the peak of B_ν expressed at the same wavelength; the two densities use different interval variables.

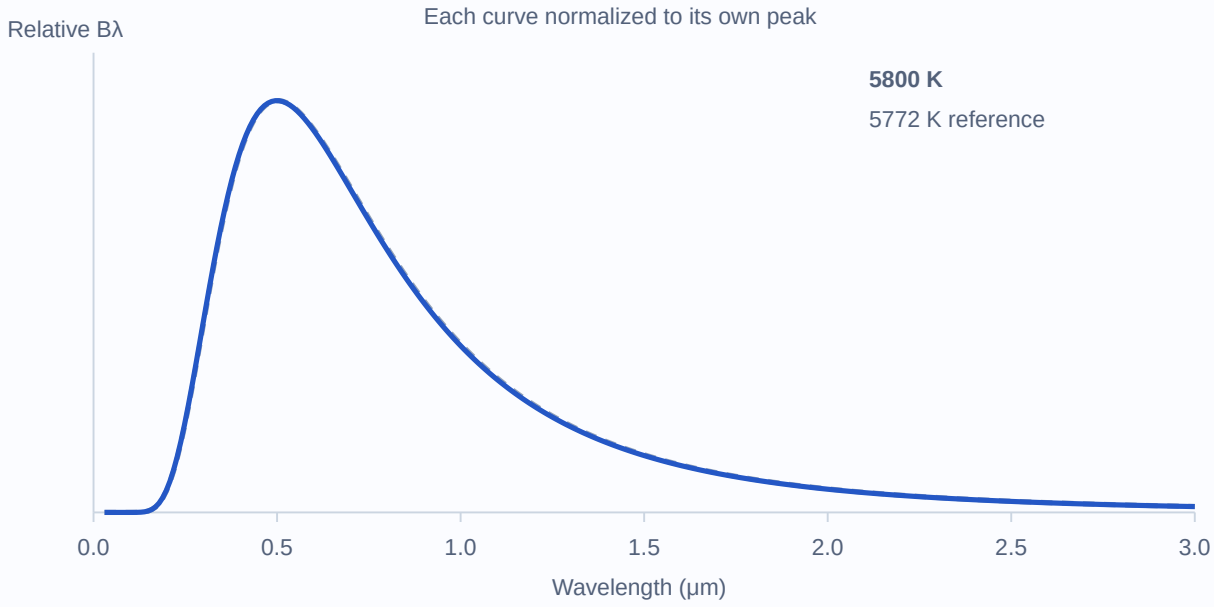
INTERACTIVE EXPERIMENT

A hotter surface is bluer—and much brighter.

Planck wavelength spectra, normalized separately for the selected star and the 5772 K reference. Radius changes luminosity but not the spectral peak.

Wien peak	Surface flux	Luminosity
0.500 μm	**6.417 × 10^7 W/m²**	**1.020 L⊙**

hc/λ_peak
2.482 eV



L/L⊙ = (R/R⊙)²(T/5772 K)⁴. The plotted photon energy is hc/λ_peak, not the mean photon energy of the spectrum.

Try this. Double the temperature. The peak wavelength halves, but the surface flux grows sixteenfold. Then double the radius at fixed temperature.

Snapshot of the experiment. Open the HTML edition to change the controls.

5. Counting photons

If a detector receives power P at wavelength λ, the ideal photon arrival rate is P/E_γ. With quantum efficiency η, the electron rate is $\eta P/E_\gamma$. For N independent detected photons, Poisson shot noise has standard deviation $\sqrt{N}$, so the photon-limited signal-to-noise ratio is $\sqrt{N}$. Doubling telescope diameter quadruples collecting area and doubles this ideal SNR for the same exposure.

6. The photoelectric threshold

A photon can eject an electron only if its energy exceeds the material's work function ϕ. Einstein's energy balance is

$$K_{\max} = hf - \phi, \qquad eV_{\text{stop}} = K_{\max}. \tag{6}$$

Increasing intensity at fixed frequency increases the number of photons, not the maximum energy of each electron. Below the threshold frequency, more intensity does not fix the energy deficit in the simple one-photon effect.

7. Radiation pushes, and dust feels it differently

Absorbed light transfers momentum at rate P/c; perfect reflection reverses the photon momentum and doubles the force. For a spherical grain, radiation pressure and stellar gravity both fall as $1/d^2$, so their ratio is independent of distance in the simple model:

$$\beta = \frac{F_{\text{rad}}}{F_{\text{grav}}} = \frac{3LQ_{\text{pr}}}{16\pi cGM\rho s}. \tag{7}$$

Here s is grain radius, ρ its density and Q_{pr} an efficiency factor. Smaller grains have larger area per unit mass and are more easily blown out. Real optics, charging and gas drag complicate the picture.

8. Choose the measurement before the formula

A spectrum's peak constrains temperature; total flux plus distance gives luminosity; luminosity and temperature give radius. Photon counts constrain detector exposure. A transit depth gives an area ratio, while an eclipse depth also depends on surface brightness at the observed wavelength.

Common traps. Use metres in Planck's and Wien's laws. Distinguish a wavelength interval from a frequency interval. Do not call $hc/\lambda_{\max}$ the average photon energy. Do not use a bolometric T^4 contrast for a narrow infrared band without checking the spectrum.

9. Lecture + Problems: predict, calculate, check

Fifteen original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Count the packets. For 500 nm light, find frequency, photon energy in eV, and the photon rate in a 1 W beam.

Worked solution →

[1] Problem 2 — Ten parsecs away. A star has $L = 4L_\odot$ and lies 10 pc away. Find its bolometric flux. Use $1\ \text{pc} = 3.0856776 \times 10^{16}$ m.

Worked solution →

[1] Problem 3 — A green-ish peak. A blackbody peaks at 500 nm in B_λ. Find its temperature and surface flux. Why need the star not look green?

Worked solution →

[2] Problem 4 — Infer a radius. A star has nine times the Sun's luminosity and twice its temperature. Find $R/R_\odot$.

Worked solution →

[2] Problem 5 — Same temperature, different size. Two stars at the same distance have equal temperature, but one has twice the radius. Find the flux ratio and magnitude difference $m_{\text{large}} - m_{\text{small}}$.

Worked solution →

[2] Problem 6 — The stopping voltage. A metal has threshold wavelength 400 nm. It is illuminated at 250 nm. Find the work function, maximum electron kinetic energy and stopping voltage in eV-based units. What changes if intensity doubles?

Worked solution →

[2] Problem 7 — A photon budget. A 2 m telescope receives 1000 photons $\text{m}^{-2}\ \text{s}^{-1}$ from a source. Total throughput is 0.25. Find detected counts and photon-limited SNR in 60 s, ignoring background. How does SNR change for a 4 m telescope?

Worked solution →

[1] Problem 8 — A solar sail. At a place where the solar flux is 1361 W/m², find radiation pressure on a perfect absorber and on a perfect reflector at normal incidence. Find the absorber force on 100 m².

Worked solution →

[2] Problem 9 — Which peak? Explain why the maximum of B_λ and the maximum of B_ν need not occur at the same numerical wavelength. Which quantity is conserved when changing variables?

Worked solution →

[1] Problem 10 — Compensating changes. A star's temperature doubles while its radius halves. Find the changes in luminosity and wavelength peak.

Worked solution →

[3] Problem 11 — From angle to temperature. A star at 100 pc has angular diameter 2 mas and luminosity $100L_\odot$. Find its radius in solar radii and effective temperature using $T_\odot = 5772$ K. Use the small-angle relation.

Worked solution →

[2] Problem 12 — Do not saturate. A pixel receives 10^5 photons/s, has quantum efficiency 0.8 and full well 50,000 electrons. Find the maximum exposure before saturation. Why can several short exposures help?

Worked solution →

[2] Problem 13 — A moving photon source. A source recedes at 300 km/s. In the low-speed limit, find the fractional wavelength and photon-energy changes. Does the photon lose energy because it slows down?

Worked solution →

[3] Problem 14 — Dust at the blowout scale. For a solar-luminosity, solar-mass star, take $Q_{\rm pr} = 1$, grain density 2000 $\mathrm{kg\,m^{-3}}$ and radius 1 μm. Evaluate β. What radius gives $\beta = 0.5$ in this model?

Worked solution →

[3] Problem 15 — An eclipse in two bands. A planet has $R_p/R_* = 0.1$, $T_p = 1500$ K and $T_* = 6000$ K. Treat both as blackbodies. Find the bolometric secondary-eclipse depth. Then find the monochromatic depth at 10 μm using Planck's law. Why are they different?

Worked solution →

Further practice and provenance

The 15 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from the papers below. The earlier club lecture compiled these official past-paper identifiers for further practice; only identifiers are reproduced, not copyrighted question statements or official solutions. Round and question labels retain the earlier compilation's notation and should be checked against the official paper.

- IOAA 2007 · TH · 1.8
- IOAA 2007 · TH · 1.9
- IOAA 2007 · TH · 1.10
- IOAA 2007 · TH · 1.11
- IOAA 2007 · TH · 1.15
- IOAA 2008 · TH · L2
- IOAA 2009 · TH · P2
- IOAA 2010 · TH · S1
- IOAA 2011 · TH · S8
- IOAA 2011 · TH · S9
- IOAA 2012 · TH · S9
- IOAA 2012 · TH · L1
- IOAA 2013 · TH · S13
- IOAA 2014 · TH · P7
- IOAA 2014 · TH · P8
- IOAA 2014 · TH · P9
- IOAA 2015 · TH · S6
- IOAA 2015 · TH · S8
- IOAA 2017 · TH · T4
- IOAA 2017 · TH · T6
- IOAA 2018 · TH · T6
- IOAA 2019 · TH · 5
- IOAA 2021 · TH · Q4
- IOAA 2022 · TH · 11
- IOAA 2023 · TH · Q2
- IOAA 2025 · TH · T04

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified.